SUMMARY

OVERVIEW

This analysis examines 2,100 customer records (after cleaning) with 2,003 recorded purchases between October 2022 and July 2025. The dataset appears synthetic: purchase amounts are spread almost uniformly between 5 and 1,000 and ratings are spread almost uniformly across 1–5. As a result, the data shows few strong patterns, and the most useful findings are about data quality and where differences are NOT present.

DATA CLEANING

- Removed an empty column, a duplicated Gender column, and Phone and Email (not useful for the analysis).
- Removed 50 fully duplicated rows (2,150 → 2,100 records).
- Standardized Gender values (Male / Female / Unknown).
- Treated invalid ages (-1 and 200) as missing instead of deleting rows or filling them with an average. Only 495 customers (about 24%) have a valid age, so age findings are limited.
- Treated ratings of 10 (291 records) as invalid, since they fall outside the 1–5 scale and no values between 6 and 9 exist.
- Treated the impossible date 32/13/2020 as missing (118 records after duplicate removal). These rows are used in all analyses except time trends.
- Kept missing product categories as "Unknown" rather than dropping them, to avoid losing a large share of purchase data.

KEY FINDINGS

Customers. Male customers make up 44.9%, female 42.4%, and 12.7% have unknown gender. Average spend is similar: 521.82 for women, 499.03 for men and 508.67 for unknown. Among customers with a known age, group averages range from 479 to 579, but each group has only about 50–70 customers (except the oldest), so these differences are not reliable.

Purchases. Total revenue is 1,021,324 and the average purchase is 509.90 (median 519.42). No outliers were found. Purchase amounts are spread evenly across all spending bands.

Product categories. Revenue is split fairly evenly: Clothing 16.0%, Electronics 15.1%, Books 14.7%, Home 14.1%, Toys 13.8%. About 26% of revenue has an unknown category, which limits category conclusions. Clothing has the highest average purchase among known categories (537).

Ratings. The average rating is 3.03 (n = 1,487 valid ratings) and each score from 1 to 5 appears about equally often. Averages by category (2.96–3.11), gender and age are all close to 3. The correlation between rating and purchase amount is about -0.03, meaning there is no relationship.

Time trends. Monthly revenue fluctuates between roughly 20K and 40K without a lasting upward or downward trend. Comparing January–July across years shows about 195K (2023), 200K (2024) and 190K (2025). The apparent dips in the first and last months and in August/September are caused by partial or fewer years of data, not real declines. Saturday has the highest average basket (550) and Tuesday–Thursday the lowest, but this difference is modest.